MUHAMMAD ZULFI HUDA AL-HAQ

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Fresh Graduate from Institut Teknologi Sepuluh Nopember (ITS) with a specialization in Automation Engineering Technology. Possessing a robust foundation in Industrial Automation systems and Electrical Design. Currently applying hands-on technical skills as a Research and Development at PT Aldzama.

Experienced in delivering impactful automation solutions from my internship as an Automation Engineer Intern at PT Flextronics Technology Indonesia, where I designed systems to automate routine tasks and simplify workflows to increase efficiency and productivity. Further expertise includes Printed Circuit Board (PCB) design and developing safety systems for electric vehicles as a former staff member of the ANARGYA ITS Team Electric Formula.

Core technical proficiencies include PLC Programming , Electro-pneumatics , Electrical Troubleshooting, and Technical Documentation. I am committed to developing impactful engineering solutions.

EDUCATION BACKGROUND

Institut Teknologi Sepuluh Nopember - Surabaya, Indonesia **Aug 2021 – Aug 2025**
Automation Engineering Technology, 3.44/4.00

SMK Negeri 1 Batam - Batam, Indonesia **Jun 2017 – Jun 2021**
Mechatronic Engineering

WORK EXPERIENCE

PT Aldzama - Gresik, Indonesia **Feb 2026 – Present**
Research and Development

- Industrial Furnace Control System Optimization : focused on enhancing combustion stability, electrical wiring efficiency, and PLC automation reliability for firebrick production processes. The project included burner optimization, PLC reprogramming and debugging, electrical panel clean-up, wiring standardization, and system validation to improve operational safety, reduce downtime, and achieve more uniform heating performance.
- QR Code-Based Tool Management System : Implemented as a Kaizen initiative to digitalize tool borrowing and inventory tracking using QR Code technology, enabling real-time monitoring, reducing manual recording errors, improving tool traceability, and increasing operational efficiency in warehouse/tool room operations.
- Remote Controlled Demolition Robot : Developed a custom remote-control system for demolition robots by reverse-engineering and modifying the original controller architecture to integrate with an in-house built control unit, reducing dependency on expensive OEM components while improving maintainability and operational flexibility.

PT Aldzama - Gresik, Indonesia **Oct 2025 – Jan 2026**
Demolition Robot Technician

- Responsible for troubleshooting and maintaining demolition robots used in industrial dismantling operations, including hydraulic, electrical, and control systems, while ensuring operational safety, equipment reliability, and minimal downtime.

PT Flextronics Technology Indonesia - Batam, Indonesia **Jan 2024 – Jun 2024**
Automation Engineer Intern

- Automatic Busbar Hipot Test : A device designed to assist operators in automatically checking insulation between six points on a busbar. It represents the application of continuous improvement, also known as KAIZEN. By automating the system, it minimizes the risk of workplace accidents.
- Wireless Auto Cut Off N2 : Developed an automatic system to turn on and off the nitrogen (N2) valve on a Surface Mount Technology (SMT) machine.
- Chamber Burner in Test : Designing and conceptualizing a chamber to test the durability of a device, by connecting it to a load for 8 hours at a temperature of 80 degrees Celsius.

- Pick to Light : Developed a pick-to-light system for order picking technology used in warehouses that makes it easier for users to find items.

PT Fast Precision Manufacturing - Batam, Indonesia **Oct 2020 – Nov 2020**

Student Intern

- Assisted in manufacturing and fabrication of contact tips.
- Supported modification and mechanical improvement of drum systems on charcoal roll machines.
- Participated in installation and assembly process of charcoal roll machines.
- Supported assembly stand roller membrane structures

ORGANIZATIONAL EXPERIENCE

Anargya ITS EV Team **Sep 2021 – Dec 2023**

Electro-Programming and Safety System Staff

- Developed the electronic control system of the electric vehicle, including sensors, control units, and safety devices
- Managing and installing wiring on an electric formula car
- Ensured all electronic and safety systems met Society of Automotive Engineers (SAE) Formula Student regulations
- Prepared and authored extensive supporting technical documents, including:
  - Failure Mode and Effects Analysis (FMEA)
  - Electrical System Form (ESF)
  - Engineering Design Supporting Documents (SUP)
  - Design Spec Sheet (DSS)
  - Software Architecture Report (SA) and Algorithm Development Report (AI)
- Served as an electronics paddock technician during Formula Student competitions

ACHIEVEMENTS

- Selected Kaizen Initiative – Kaizen program for February period (2026)
- 4th position Photovoltaic Design Competition 2.0 Issued by Himasiskal ITS X PT Hasnur Internasional Shipping (2023)
- 3rd position HVL National Competition Issued by High Voltage Laboratory ITS Surabaya (2022)
- Finalist of PLN Innovation & Competition in Electricity (PLN ICE) 2022 in Electrifying Agriculture category Issued by PLN Puslitbang (2022)
- Runner-up of Overall Result at 6th Pi-EV Formula Student Concept Challenge held by Formula Bharat India Formula Student Competition (2022)
- 2nd winner in selection of Training Center for 13th Asean Skill Competition in Skill of CNC Maintenance and Repair (2020)
- 2nd winner the national selection for 13th Asean Skill Competition in Skill of CNC Maintenance and Repair (2019)

PROJECTS

- Final Project : Designing a cooling system based on fuzzy logic controller to increase energy efficiency while maintaining the ideal temperature of the motor controller (2025)
- Designed and developed a remote-controlled mini boat for oil spill detection under an industrial project collaboration with PT Pertamina Trans Kontinental (2023)
- Create and design the concept of ship that can be controlled via remote control by technicians to isolate oil spills (2023)
- Prototype of Sorting Machine with Qr Reader in the logistics industry (2022)